

GIT and GITHUB

Step 1: Download and Install Git

Go to  <https://git-scm.com/downloads>

Choose Windows (or Mac/Linux depending on your OS).

Run the installer → click Next for most defaults.

Keep Git Bash option selected.

Choose your preferred editor (e.g., Notepad++ or VS Code).

Finish installation.

Step 2: Verify Git Installation

Open Git Bash and run:

```
git --version
```

 You should see something like:

```
git version 2.46.0.windows.1
```

Step 3: Configure Git User Info (one-time setup)

Tell Git who you are — these details appear in commits.

```
git config --global user.name "Your Name" [GITHUB UserName]
```

```
git config --global user.email your\_email@example.com [GITHUB EMailID]
```

Verify configuration:

```
git config --list
```

Step 4: Create or Open a Local Project Folder

Example: Open Bash and Redirect to Local Project Folder

```
cd "D:\GIT and GITHUB\GIT Project-2"
```

Before: HP@DESKTOP-DCUKDOL MINGW64 ~

```
$ cd "D:\GIT and GITHUB\GIT Project-2"
```

After: HP@DESKTOP-DCUKDOL MINGW64 /d/GIT and GITHUB/GIT Project-2

Step 5: Initialize a Git Repository

First Check it created folder is normal folder or Repo

```
$ ls -a
```

It shows only like below it is only normal folder now covert to repo

```
./ ../
```

```
$ git init
```

✓ You'll see:

Initialized empty Git repository in D:/GIT and GITHUB/GIT Project-2/.git/

```
$ ls -a
```

```
./ ../ .git/ [Now it is repo]
```

Step 8: Create a Repository on GitHub

Go to <https://github.com>

Log in to your account.

Click New Repository (+ sign → New repository).

Enter a repo name (e.g., SampleGIT).

Choose Public or Private.

Don't initialize with a README (since you already have one locally).

Click Create repository.

You'll get a URL like:

https://github.com/krishnasoftwaretrainer/QT_COREJAVA_225BATCH.git

Step 9: Connect Local Repo to GitHub

Back in Git Bash:

```
git remote add origin https://github.com/krishnasoftwaretrainer/QT\_COREJAVA\_225BATCH.git
```

[GitHub Repo URL]

Verify it:

```
git remote -v
```

✓ **Should show:**

```
origin https://github.com/krishnasoftwaretrainer/QT\_COREJAVA\_225BATCH.git (fetch)
```

```
origin https://github.com/krishnasoftwaretrainer/QT\_COREJAVA\_225BATCH.git (push)
```

Note: Sometimes pointing to your old GitHub repo.

Change the remote URL or Remove old origin and add new one.

Option 1: **git remote set-url origin**

https://github.com/krishnasoftwaretrainer/QT_COREJAVA_225BATCH.git

Then check:

bash

git remote -v

Option 2: **git remote remove origin**

Then add the new one:

bash

git remote add origin

https://github.com/krishnasoftwaretrainer/QT_COREJAVA_225BATCH.git

Step 10: Push Local Repo to GitHub (Detailed Flow)

◆ **1 Local Files (Working Directory)**

This is where your actual project files live on your computer.

📌 Example:

D:/GIT and GITHUB/GIT Project-1/

When you edit or create new files, Git sees them as untracked or modified.

✅ Check their status:

git status

✖ Output Example:

Untracked files:

(use "git add <file>..." to include in what will be committed)

index.html

style.css

◆ **2 Staging Area (Index)**

The staging area holds files you're preparing to commit.

✅ Add all files to staging:

git add .

Check again:

git status

✖ Output Example:

Changes to be committed:

(use "git rm --cached <file>..." to unstage)

new file: index.html

new file: style.css

◆ 3 Commit Changes (Local Repository)

Once your files are staged, you commit them — this saves a version snapshot in your local Git database.

✓ Commit command:

```
git commit -m "Initial commit"
```

✖ Output Example:

```
[master (root-commit) abc1234] Initial commit
```

```
2 files changed, 50 insertions(+)
```

```
create mode 100644 index.html
```

```
create mode 100644 style.css
```

✓ Check commit history:

```
$ git log
```

```
commit b9e2ebc78632c34d0c26459992b7676b8130aaba (HEAD -> master, origin/master)
```

```
Author: krishnasoftwaretrainer <krishnasoftwaretrainer@gmail.com>
```

```
Date: Fri Nov 21 16:00:00 2025 +0530
```

Upto Java Tokens

◆ 4 Push to GitHub (Remote Repository)

Now that your changes are safely committed locally, push them to your GitHub repo.

Add GitHub Remote (only once):

```
git remote add origin https://github.com/QAKRISHNAN/SampleGIT.git
```

[error: remote origin already exists.] If you are done already.

Push Command:

If your branch is master:

```
git push -u origin master
```

If your branch is main:

```
git push -u origin main
```

✖ Output Example:

info: please complete authentication in your browser...

Enumerating objects: 24, done.

Counting objects: 100% (24/24), done.

Delta compression using up to 8 threads

Compressing objects: 100% (20/20), done.

Writing objects: 100% (24/24), 5.27 KiB | 224.00 KiB/s, done.

Total 24 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)

To https://github.com/krishnasoftwaretrainer/QT_COREJAVA_225BATCH.git

* [new branch] master -> master

branch 'master' set up to track 'origin/master'.

❏ Step 11: Verify on GitHub

Go to your GitHub repo URL:

👉 <https://github.com/QAKRISHNAN/SampleGIT>

✅ You should now see all your project files uploaded successfully!

⚙️ Optional Useful Commands

Command	Purpose
git log	Shows commit history
git branch	Lists all branches
git checkout -b branch_name	Creates and switches to a new branch
git pull origin master	Pulls latest changes from GitHub
git push origin master	Pushes changes to GitHub

How to pull (or clone) GitHub repository to your local system.

✂️ Step-by-Step: Pull / Clone a GitHub Repository to Local

◆ 1. Copy the GitHub Repository URL

Go to the GitHub repo page (for example):

👉 <https://github.com/SomeUser/SomeProject.git>

Click Code → HTTPS → Copy URL

The URL will look like:

<https://github.com/username/repository.git>

◆ 2. Open Git Bash in the Folder You Want to Clone Into

[Create new folder or Use existing folder]

Example: `cd "/d/GIT and GITHUB/"`

◆ 3. Clone the Repository

Run:

```
git clone https://github.com/username/repository.git
```

✿ Example:

```
git clone https://github.com/QAKRISHNAN/SampleGIT.git
```

✓ Output:

Cloning into 'SampleGIT'...

remote: Enumerating objects: 25, done.

remote: Counting objects: 100% (25/25), done.

Receiving objects: 100% (25/25), done.

Resolving deltas: 100% (5/5), done.

◆ 4. Check Remote (GitHub) Connection

```
git remote -v
```

✓ Output:

```
origin https://github.com/QAKRISHNAN/SampleGIT.git (fetch)
```

```
origin https://github.com/QAKRISHNAN/SampleGIT.git (push)
```

◆ 5. Pull Latest Updates Later

Once cloned, you can always pull new changes from the GitHub repo using:

```
$ git pull origin master
```

or

```
$ git pull origin main
```

(depends on the branch name — check using `git branch -a`)

✓ Output:

Already up to date.

or

Updating 2a4c12f..b6d3f12

Fast-forward

index.html | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)

✿ Complete Git Command Sequence — Local ↔ GitHub

Step No.	Git Command	Description / Purpose
1	git	Checks if Git is installed and available.
2	cd "D:\GIT and GITHUB\GIT Project-2"	Navigates to your project directory.
3	git status	Displays current repo status (untracked or modified files).
4	git init	Initializes a new local Git repository in the current folder.
5	ls -a	Lists all files (including hidden .git folder).
6	git remote add origin https://github.com/QAKRI SHNAN/JavaRepo.git	Links local repo with remote GitHub repository.
7	git remote -v	Verifies the remote GitHub link for fetch & push.
8	git add .	Adds (stages) all files from local folder to staging area.
9	git commit -m "Initial commit"	Commits staged files with a message to local repository.
10	git log	Shows commit history (commit ID, author, date, message).
11	git push -u origin main	Pushes committed changes from local main branch to GitHub.
12	git push -u origin master	(Alternative) Pushes changes to master branch if your repo uses it.
13	git clone "https://github.com/nerdseker365/TEXT_BOOKS-AND-MATERIALS.git"	Clones an existing GitHub repo into your local system.
14	git remote -v	Checks the remote URLs for the cloned repository.
15	git pull origin master	Pulls latest changes from remote master branch into local repo.

✓ Summary of Workflow

Category	Command Examples	Purpose
Initialization	git init, git status	Set up a new local Git repository.
Remote Setup	git remote add origin <URL>	Connect local repo to GitHub.

Staging & Committing	git add ., git commit -m "message"	Prepare and record local changes.
Push to GitHub	git push -u origin master	Upload changes to GitHub repository.
Clone Repo	git clone <URL>	Copy GitHub repo to local machine.
Pull Updates	git pull origin master	Sync latest changes from GitHub.

Git & GitHub Commands Cheat Sheet

Prepared by Krishna N (QA Engineer)

Step No.	Git Command	Description / Purpose
1■■■	git	Checks if Git is installed and available.
2■■■	cd "D:\GIT and GITHUB\GIT Project-2"	Navigates to your project directory.
3■■■	git status	Displays current repo status (untracked or modified files).
4■■■	git init	Initializes a new local Git repository in the current folder.
5■■■	ls -a	Lists all files (including hidden .git folder).
6■■■	git remote add origin <repo URL>	Links local repo with remote GitHub repository.
7■■■	git remote -v	Verifies the remote GitHub link for fetch & push.
8■■■	git add .	Adds all files from local folder to staging area.
9■■■	git commit -m "Initial commit"	Commits staged files with a message to local repository.
■■■	git log	Shows commit history (commit ID, author, date, message).
1■■■1■■■	git push -u origin main	Pushes committed changes from local main branch to GitHub.
1■■■2■■■	git push -u origin master	Pushes committed changes from local master branch to GitHub.
1■■■3■■■	git clone <repo URL>	Clones an existing GitHub repo into your local system.
1■■■4■■■	git remote -v	Checks the remote URLs for the cloned repository.
1■■■5■■■	git pull origin master	Pulls latest changes from remote master branch into local repo.

Git Workflow Diagram & Explanations

Prepared by Krishna N (QA Engineer)

This guide explains the flow of how files move through different Git stages — from Local Files to Staging Area, then to Local Repository, and finally to the Remote GitHub Repository. It also includes Clone and Pull operations for synchronization.

LOCAL FILES (Working Directory) → **STAGING AREA (Index)** → **LOCAL REPOSITORY (Commit)** → **REMOTE REPOSITORY (GitHub)**

Stage	Purpose	Git Commands
1■■■ Local Files	Your working project files on your system.	git status
2■■■ Staging Area	Files selected for the next commit.	git add .
3■■■ Local Repository	Your committed code stored locally.	git commit -m "message"
4■■■ Remote Repository	Your project hosted on GitHub (shared repo).	git push -u origin master / main
■■■ Clone	Copy an existing GitHub repo to your local system	git clone <repo_url>
■■■ Pull	Fetch and merge changes from GitHub to local repo	git pull origin master / main

◆ 1. Staging (Adding to Staging Area)

💡 What it means:

When you make changes in your local files (edit, add, delete), Git first sees them as untracked or modified.

Before saving those changes permanently in the local repository, you must stage them — i.e., mark which changes you want to include in the next commit.

🧠 Think of it like:

Selecting which files you want to pack into your next “snapshot” (commit).

✚ **Command:**

```
git add <filename>    # stage a single file
```

```
git add .              # stage all changes in the current directory
```

📌 Example:

```
git add index.html
```

```
git add src/           # add all files in src folder
```

After this step, files are moved to the Staging Area.

◆ 2. Commit

💡 What it means:

A commit takes all the staged changes and saves a snapshot of them in your local Git repository (on your computer).

It's like saving a checkpoint in your project history.

🧠 Think of it like:

Taking a photo of your project at that moment (after you've staged the changes).

✖ Command:

```
git commit -m "Added new login feature"
```

This step:

Stores changes in your local repository.

Adds a unique commit ID (SHA).

Doesn't affect GitHub yet — still local.

◆ 3. Push

💡 What it means:

A push sends your local commits to the remote repository (like GitHub, GitLab, or Bitbucket).

Now your team (or anyone with access) can see your changes.

🧠 Think of it like:

Uploading your saved snapshots from your computer to the cloud (GitHub).

✖ Command:

```
git push origin main
```

or (if your branch name is master)

```
git push origin master
```

-+



Difference Between Staging, Commit, and Push in Git

Staging (git add)	Commit (git commit)	Push (git push)
To prepare or select changes that you want to include in the next commit	To prepare or select changes that you want to include in the next commit	To prepare or select changes that you want to include in the next commit
git add <file> or git add .	git commit -m "message"	git push origin <branch>

In the Staging Area (Index)	In the Local Repository	In the Remote Repository
Prepares files for commit but doesn't save permanently	Saves snapshot (commit) locally	No local change (just uploads)
Yes, you can stage multiple times before committing	Yes, you can make multiple commits before pushing	Yes, you can push multiple commits at once
Working Directory → Staging Area	Staging Area → Local Repository	Local Repository → Remote Repository